MATRIXPOLICY FOR ROLE-AWARE OPTIMIZATION OF RATIONAL TRANSFORMER NONLINEARITIES

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ABSTRACT

A useful LLM pretraining recipe moves the loss-time frontier by reaching the same validation loss with fewer tokens or less wall-clock time. We target this objective through the Transformer nonlinear sublayer, treating it as an optimizer-visible interface. MatrixPolicy with Rational Latent Basis (RLB) replaces a fixed scalar response with grouped rational responses that expose local gain, coefficient-activity, and matrix-balance statistics to a role-aware update. MatrixPolicy uses these signals to update only the sublayer input and output matrices, while rational coefficients and the rest of the Transformer remain on AdamW. Across DCLM, FineWeb-Edu, FineWeb, Dolma-sample, and C4 with a fixed 12-layer, width-768 decoder, MatrixPolicy reaches matched validation-loss targets with fewer tokens and lower measured arrival time than SiLU controls under the same protocol. It saves 19.5–22.8% target-arrival tokens versus SiLU with AdamW and 29.2–34.0% versus SiLU with Muon at 100M tokens, and 22.3–26.4% and 6.7–9.8%, respectively, at 300M tokens. [todo: Insert the larger-scale training result here.]

1 INTRODUCTION

Recent LLM optimizer papers frame practical efficiency through loss-time and compute-time frontiers, asking which training recipe reaches a target loss with fewer tokens, FLOPs, or wall-clock time (Shah et al., 2025; Liu et al., 2025; Semenov et al., 2025). This criterion also exposes an activation-design question. In a modern decoder block, the nonlinear sublayer is a learned matrix–nonlinearity–matrix map. The activation sets local gain and curvature, while the adjacent matrices choose and recombine the coordinates on which that response acts. Standard training pipelines leave this local structure implicit to the optimizer. Activation coefficients and adjacent matrices are updated as ordinary parameters, leaving gain, coefficient-activity, and role information inside the sublayer unused by the matrix update.

We make this interface concrete with Rational Latent Basis (RLB) and MatrixPolicy. RLB maps residual states into grouped latent coordinates, normalizes each group by its RMS radius, and applies a trainable real-pole-free P5/Q4 rational response. The resulting response-gain, coefficient-activity, matrix-pressure, and pair-balance signals are detached from the forward graph and passed to MatrixPolicy. MatrixPolicy uses them to update only the RLB input and output matrices; attention, embeddings, normalization parameters, rational coefficients, and ordinary Transformer tensors remain on AdamW. This local division lets the nonlinear sublayer expose the quantities it creates without changing the optimizer used for the rest of the model.

We evaluate target arrival by measuring the tokens and wall-clock time required to reach a fixed validation-loss target under the same model, corpus, and protocol. In the completed study, the same 12-layer, width-768 causal Transformer is trained on DCLM, FineWeb-Edu, FineWeb, Dolma-sample, and C4. The experiments include RLB-only controls and matched SiLU with AdamW and SiLU with Muon baselines under the same model, data, and measurement protocol. MatrixPolicy reaches shared validation-loss targets earlier at both 100M and 300M token budgets. [todo: Insert the larger-scale training result here.] The contribution is an activation design that exposes optimizer-visible matrix roles, a localized role-aware optimizer for the exposed matrix pair, and matched evidence that this interface reduces target-arrival tokens and wall-clock time across five pretraining corpora.

2 Related Work

Activation design for Transformer nonlinear sublayers has moved from fixed pointwise curves to gated matrix sublayers. Early Transformer-style sublayers used fixed pointwise responses such as GELU or Swish/SiLU (Vaswani et al., 2017; Hendrycks & Gimpel, 2016; Ramachandran et al., 2017). GLU introduced multiplicative feature selection for language modeling (Dauphin et al., 2017), GLU variants such as SwiGLU improved Transformer sublayers in compute-matched settings (Shazeer, 2020), and large decoder models adopted SwiGLU together with pre-normalized Transformer components (Chowdhery et al., 2022; Touvron et al., 2023). These works identify the nonlinear block as the natural comparison point for RLB because the response is coupled to the matrices that prepare and mix it. RLB keeps that sublayer location while replacing the fixed curve with grouped trainable rational responses.

Choosing a trainable rational response builds on earlier evidence that low-degree quotients can serve as learnable nonlinearities. Padé Activation Units learned such quotients end to end as activation functions (Molina et al., 2020), rational neural networks gave approximation efficiency results for smooth-function classes (Boullé et al., 2020), and RAFT evaluated rational activation functions inside Transformer pretraining and fine-tuning (Fang et al., 2023). Recent theory strengthens the motivation by proving expressivity advantages for trainable low-degree rational activations over a broad set of modern fixed activations, including gated and Transformer-style nonlinearities (Tang & Townsend, 2026). RLB therefore follows the rational-activation lineage; MatrixPolicy tests the additional effect of making the surrounding matrices optimizer-visible.

Making the surrounding matrices optimizer-visible connects RLB to a fast-moving LLM optimizer literature. Adam and AdamW remain central baselines (Kingma & Ba, 2015; Loshchilov & Hutter, 2019), while state-efficient or diagonal adaptive methods such as Adafactor and CAME reduce optimizer memory or alter adaptive-state estimates (Shazeer & Stern, 2018; Luo et al., 2023). Other recent methods change the update rule itself. Lion, Schedule-Free methods, and cautious optimizers alter momentum, sign, or schedule behavior (Chen et al., 2023; Defazio et al., 2024; Liang et al., 2026), while Sophia, GaLore, and Adam-mini change curvature, low-rank, or per-block learning-rate structure (Liu et al., 2024; Zhao et al., 2024; Zhang et al., 2025). Practical-efficiency studies evaluate optimizers by the tokens, compute, or time needed to reach matched losses (Shah et al., 2025; Schlotthauer et al., 2025; Semenov et al., 2025; Wen et al., 2026). The matrix restriction in MatrixPolicy connects most directly to matrix-aware optimization. Shampoo uses tensor preconditioners (Gupta et al., 2018), SOAP stabilizes Shampoo-style preconditioning by running Adam in the preconditioner eigenbasis (Vyas et al., 2025), and Muon uses orthogonalized matrix updates for LLM training (Liu et al., 2025). Recent optimizer comparisons emphasize fair tuning, model-scale dependence, and setting-dependent tradeoffs (Zhao et al., 2025; Wen et al., 2026). MatrixPolicy follows this matrix-aware direction, with updates restricted to the RLB input/output matrix pair and conditioned on activation-derived sublayer statistics.

3 Method

RLB and MatrixPolicy are designed as a single interface. RLB defines the nonlinear sublayer and the detached summaries it exposes; MatrixPolicy consumes those summaries to update the two adjacent matrices. Rational coefficients and ordinary Transformer parameters remain on the AdamW path in the evaluated configuration.

Throughout the method we use

$$\mathrm{clip}(x, a, b) = \min\{\max\{x, a\}, b\}, \tag{1}$$

$$\mathrm{smoothstep}(a, b, x) = 3\omega^2 - 2\omega^3, \qquad \omega = \mathrm{clip}\left(\frac{x-a}{b-a}, 0, 1\right). \tag{2}$$

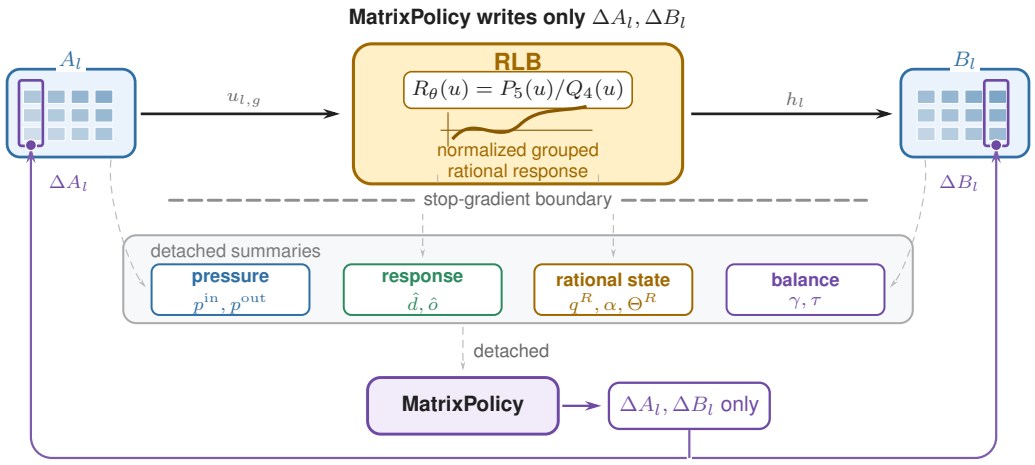

Figure 1: Activation–optimizer interface for RLB. In layer l, A_l maps the sublayer input into normalized rational groups and B_l recombines the rational features $h_l = \text{concat}_{g=1}^{G} h_{l,g}$. Forward RLB statistics, gradient pressure/activity EMAs, and pair-balance measurements cross a stop-gradient boundary as policy inputs. MatrixPolicy reads these summaries and writes only ΔA_l and ΔB_l; rational coefficients are AdamW-trained and policy-observed.

3.1 Rational Latent Basis

For layer $l \in \{1, \dots, L\}$, let $x_l \in \mathbb{R}^d$ be the input to the nonlinear sublayer. RLB uses an input matrix $A_l \in \mathbb{R}^{H \times d}$ and an output matrix $B_l \in \mathbb{R}^{d \times H}$ with latent width $H = Gm$. Let $A_{l,g} \in \mathbb{R}^{m \times d}$ denote the row block of A_l, and let $B_{l,g} \in \mathbb{R}^{d \times m}$ denote the matching column block of B_l. In the implementation, A_l and B_l are the bias-free `in_proj.weight` and `out_proj.weight` of the nonlinear sublayer. The evaluated runs use an RMS floor $\epsilon_{\text{rms}} = 10^{-6}$. Each group has a trainable P5/Q4 rational response

$$P_{l,g}(u) = \sum_{i=0}^{5} a_{l,g,i} u^i, \tag{3}$$

$$Q_{l,g}(u) = 1 + |b_{l,g,1}|\,|u| + |b_{l,g,2}|\,u^2 + |b_{l,g,3}|\,|u|^3 + |b_{l,g,4}|\,u^4, \tag{4}$$

$$R_{l,g}(u) = P_{l,g}(u)/Q_{l,g}(u). \tag{5}$$

Define the preactivation, RMS radius, normalized coordinate, and grouped rational feature by

$$z_{l,g}(x_l) = A_{l,g} x_l, \qquad \rho_{l,g}(x_l) = \sqrt{m^{-1}\|z_{l,g}(x_l)\|_2^2 + \epsilon_{\text{rms}}}, \tag{6}$$

$$u_{l,g}(x_l) = \frac{z_{l,g}(x_l)}{\rho_{l,g}(x_l)}, \qquad h_{l,g}(x_l) = \rho_{l,g}(x_l) R_{l,g}(u_{l,g}(x_l)), \qquad h_l = \text{concat}_{g=1}^{G} h_{l,g}. \tag{7}$$

The RLB forward map is the single block expression

$$y_l = B_l h_l = \sum_{g=1}^{G} B_{l,g} \rho_{l,g}(x_l) R_{l,g}\left(\frac{A_{l,g} x_l}{\rho_{l,g}(x_l)}\right). \tag{8}$$

The scalar functions $P_{l,g}$, $Q_{l,g}$, and $R_{l,g}$ are applied coordinatewise when their input is a vector. Since $Q_{l,g}(u) \geq 1$ for all real u, the implemented response has no real pole. The response is piecewise rational because of the absolute-value terms; derivative statistics use the autodiff convention $\text{sign}(0) = 0$ and equations involving $R'_{l,g}$ are interpreted almost everywhere. Numerator and denominator coefficients are initialized by a SiLU fit on $[-5, 5]$ and are then trained by AdamW. The evaluated model uses only these groupwise P5/Q4 rational responses.

The forward map in equation 8 separates the normalized coordinate at which the rational response is evaluated from the group radius returned to the residual stream. This radius-factor form exposes

the positive homogeneity behind the matrix-pair coordinate formalized in Appendix B. The floor $\epsilon_{\mathrm{rms}} > 0$ is an additive squared-RMS stabilizer. For fixed curves $R_l = \{R_{l,g}\}_{g=1}^{G}$, write this block map as $f_l(x; A_l, B_l, R_l)$. For any fixed input x such that $A_{l,g}x \neq 0$ for every group, if $\epsilon_{\mathrm{rms}} = 0$, then for any $\lambda \in \mathbb{R}^{G}_{>0}$ and $D_l(\lambda) = \mathrm{diag}(\lambda_1 I_m, \ldots, \lambda_G I_m)$,

$$f_l(x; A_l, B_l, R_l) = f_l(x; D_l(\lambda)A_l, B_l D_l(\lambda)^{-1}, R_l). \tag{9}$$

This positive matrix-pair rescaling is exact only in the unfloored map. In the actual training dynamics with the stabilizing RMS floor and optimizer state, MatrixPolicy uses the same direction as a bounded balance coordinate.

3.2 MatrixPolicy

MatrixPolicy is the optimizer-side policy attached to the RLB matrix pair (A_l, B_l). We use the name for the matrix-specific controls in the evaluated optimizer configuration. Other Transformer tensors follow their assigned backbone optimizer, and the rational coefficients contribute activity signals while retaining their AdamW no-decay updates in the reported runs. The policy reads detached summaries. These summaries are produced by the RLB forward hook, by post-global-clipping gradients observed before MatrixPolicy rescales them, and by fixed-grid probes of the rational response. They feed, in order, group-gradient gating, a role/depth matrix step, and reciprocal pair rescaling.

This ordering matters. Let t be the one-based optimizer step after the wrapper increments its counter, T the planned number of steps, and $\xi_t = t/T$. The surrounding optimizer wrapper first constructs detached gradient memories from the current step; the inner matrix optimizer uses those memories to rescale matrix gradients and apply AdamW/Muon updates; the wrapper then measures the updated matrix pair and applies the reciprocal rescale. No summary tensor enters the forward graph. For notation, set $M_{l,\mathrm{in}} = A_l$ and $M_{l,\mathrm{out}} = B_l$, with $\sigma \in \{\mathrm{in}, \mathrm{out}\}$ denoting the two roles. For any tensor V, define

$$\mathrm{rms}(V) = \sqrt{\mathrm{mean}(V^2)}, \qquad \mathrm{rms}_\epsilon(V) = \sqrt{\mathrm{mean}(V^2) + \epsilon}. \tag{10}$$

The evaluated implementation uses $\epsilon_{\mathrm{live}} = 10^{-6}$ for cached forward gains and $\epsilon_p = \epsilon_{\mathrm{probe}} = 10^{-8}$ for pressure, matrix-ratio, and fixed-grid probe floors.

The first family of summaries is the on-policy pressure and activity memory. Here "on-policy" means that the wrapper records statistics from the current training trajectory. All gradients in this paragraph are optimizer-visible loss gradients after global clipping and before MatrixPolicy group-gradient scaling. Let $\Theta^R_{l,g}$ collect the trainable rational-response coefficient blocks in group (l, g); for the reported P5/Q4 response these are the numerator and denominator coefficient tensors. The count $|\Theta^R_{l,g}|$ is a block count, and each mean is over scalar entries in one block. The instantaneous detached measurements are

$$p^{\mathrm{in}}_{l,g} = \frac{\mathrm{rms}_{\epsilon_p}(\nabla A_{l,g})}{\mathrm{rms}_{\epsilon_p}(A_{l,g})}, \qquad p^{\mathrm{out}}_{l,g} = \frac{\mathrm{rms}_{\epsilon_p}(\nabla B_{l,g})}{\mathrm{rms}_{\epsilon_p}(B_{l,g})}, \tag{11}$$

$$p^{R}_{l,g} = \left[\frac{1}{|\Theta^R_{l,g}|} \sum_{\theta \in \Theta^R_{l,g}} \mathrm{mean}((\nabla\theta)^2) + \epsilon_p \right]^{1/2}. \tag{12}$$

The matrix quantities are relative RMS-gradient proxies, while $p^R_{l,g}$ is raw coefficient-gradient energy. The wrapper stores detached EMA memories q^{in}, q^{out}, and q^R,

$$q^r_{l,g,t} = 0.95\, q^r_{l,g,t-1} + 0.05\, p^r_{l,g,t}, \qquad r \in \{\mathrm{in}, \mathrm{out}, R\}, \tag{13}$$

initialized from the first observed values. We reserve q in this subsection for these detached policy memories; the rational denominator remains denoted by $Q_{l,g}$. The two contrasts used by all three controls are

$$\pi_{l,g} = \log q^{\mathrm{in}}_{l,g} - \log q^{\mathrm{out}}_{l,g}, \qquad \alpha_{l,g} = \log q^R_{l,g} - \frac{1}{2}\left(\log q^{\mathrm{in}}_{l,g} + \log q^{\mathrm{out}}_{l,g}\right). \tag{14}$$

The pressure contrast $\pi_{l,g}$ compares the two adjacent matrix roles, and the activity contrast $\alpha_{l,g}$ compares rational-coefficient gradient energy with the geometric mean of those role memories. Appendix C.1 derives the local gradient geometry behind these proxies. We drop the step subscript when the time is clear.

The second family of summaries comes from the RLB forward-statistics hook. The hook samples normalized coordinates and caches per-group response and derivative RMS values,

$$\hat{d}^{\text{live}}_{l,g} = \sqrt{\text{mean}\Big(R'_{l,g}(u_{l,g})^2\Big) + \epsilon_{\text{live}}}, \qquad \hat{o}^{\text{live}}_{l,g} = \sqrt{\text{mean}(R_{l,g}(u_{l,g})^2) + \epsilon_{\text{live}}}. \tag{15}$$

The mean is over sampled scalar normalized coordinates for group (l, g) across the most recent refreshed sample, sequence positions, and the m group coordinates. The derivative gain is the input-role gain proxy, and the response gain is the output-role gain proxy. If a cache is unavailable at a step, the implementation sets the corresponding gain multiplier to identity while keeping the pressure/activity memories active.

The third family is produced only for the post-step pair rescale. Let $\mathcal{U} = \{u_k\}_{k=1}^{129}$ be a uniform probe grid over $[-5, 5]$, and define $\text{rms}_{\mathcal{U}}(\varphi) = \sqrt{|\mathcal{U}|^{-1} \sum_{u_k \in \mathcal{U}} \varphi(u_k)^2}$. The fixed-grid probe producer refreshes every ten optimizer steps and emits

$$\hat{d}^{\text{probe}}_{l,g} = \sqrt{\text{rms}_{\mathcal{U}}(R'_{l,g})^2 + \epsilon_{\text{probe}}}, \qquad \hat{o}^{\text{probe}}_{l,g} = \sqrt{\text{rms}_{\mathcal{U}}(R_{l,g})^2 + \epsilon_{\text{probe}}}. \tag{16}$$

These probes describe rational curve shape on a shared interval; they are separate from the sampled live gains in equation 15. After the role/depth matrix step and immediately before reciprocal rescaling, the wrapper also measures the detached pair-state coordinate

$$\gamma_{l,g} = \log \text{rms}_{\epsilon_p}(A_{l,g}) - \log \text{rms}_{\epsilon_p}(B_{l,g}). \tag{17}$$

Only the probe gains and $\gamma_{l,g}$ are emitted by this summary family. The third control then computes $\tau_{l,g}$ and $\chi^{\text{rs}}_{l,g}$ from those summaries and from the pressure and activity memories.

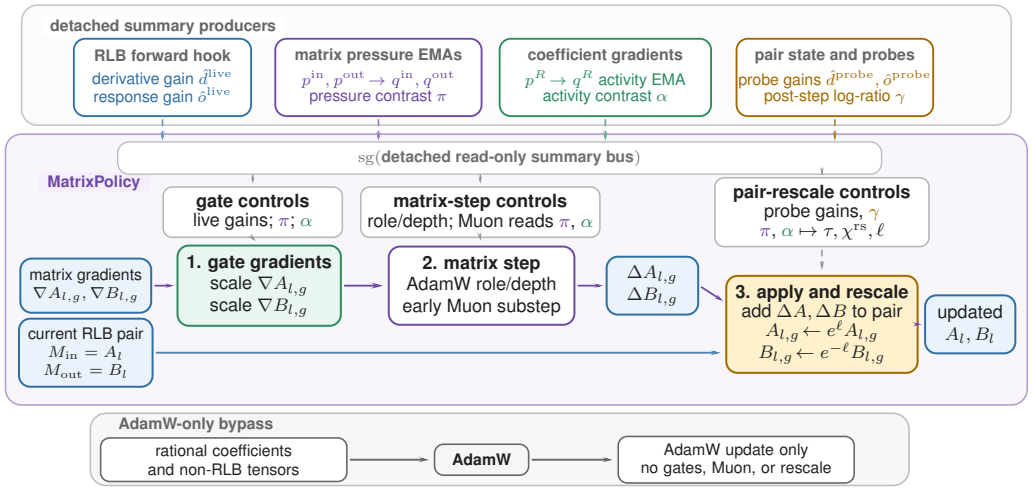

Figure 2: Detached-summary route for the evaluated MatrixPolicy system. The top band produces stop-gradient summaries from forward hooks, pre-policy gradients, fixed-grid probes, and post-step pair-state measurements; the read-only bus routes them to three MatrixPolicy consumers. The lower policy band separates the gradient/update lane from the current matrix-pair state lane, while rational coefficients and non-RLB tensors remain AdamW-only.

3.2.1 Group-Gradient Gating

The first control redistributes gradient scale among groups inside each matrix role. It reads the sampled live gains, the pressure contrast $\pi_{l,g}$, and the activity contrast $\alpha_{l,g}$, then produces scaled gradients for A_l and B_l. A smooth gate $\phi_t = \text{smoothstep}(0.02, 0.30, \xi_t)$ turns on this control in the reported runs. The evaluated manifest sets gain, pressure, and activity strengths to 0.20,

0.10, and 0.20, and clips group multipliers to $[0.75, 1.35]$. For positive group values v_j, define $\mathrm{geomean}_{j=1}^{G}(v_j) = \exp\Big(G^{-1}\sum_{j=1}^{G}\log v_j\Big)$. The group-gradient rule is

$$\begin{aligned}
\tilde{\pi}_{l,g} &= \mathrm{clip}(\pi_{l,g}, -1.50, 1.50), \qquad c_{l,g}^{\mathrm{activity}} = \exp\left[-0.20\phi_t\,\mathrm{ReLU}\left(\frac{\alpha_{l,g}-0.05}{0.45}\right)\right], \\
c_{l,g,\mathrm{in}}^{\mathrm{gain}} &= \left(\frac{\mathrm{geomean}_{j=1}^{G}\,\hat{d}_{l,j}^{\mathrm{live}}}{\hat{d}_{l,g}^{\mathrm{live}}}\right)^{0.20\phi_t}, \; c_{l,g,\mathrm{out}}^{\mathrm{gain}} = \left(\frac{\mathrm{geomean}_{j=1}^{G}\,\hat{o}_{l,j}^{\mathrm{live}}}{\hat{o}_{l,g}^{\mathrm{live}}}\right)^{0.20\phi_t}, \\
c_{l,g,\mathrm{in}}^{\mathrm{pressure}} &= \exp(-0.10\phi_t\tilde{\pi}_{l,g}), \qquad c_{l,g,\mathrm{out}}^{\mathrm{pressure}} = \exp(+0.10\phi_t\tilde{\pi}_{l,g}), \\
c_{l,g,\sigma}^{\mathrm{raw}} &= c_{l,g,\sigma}^{\mathrm{gain}}c_{l,g,\sigma}^{\mathrm{pressure}}c_{l,g}^{\mathrm{activity}}, \qquad c_{l,g,\sigma} = \mathrm{clip}\left(\frac{c_{l,g,\sigma}^{\mathrm{raw}}}{\mathrm{geomean}_{j=1}^{G}\,c_{l,j,\sigma}^{\mathrm{raw}}}, 0.75, 1.35\right).
\end{aligned} \tag{18}$$

Live derivative gain controls the input role, live response gain controls the output role, and pressure acts with opposite signs on the two roles. When $\pi_{l,g} > 0$, the input-side relative-gradient EMA is larger than the output-side EMA, so the gate reduces the input-role gradient scale and increases the output-role gradient scale for that group. The activity term damps groups whose rational coefficient gradients are large relative to the adjacent matrix memories. The geometric recentering preserves the role-wise mean log multiplier before clipping; Appendix C.2 gives the log-space identity.

The scaled matrix gradients are

$$\widetilde{\nabla}A_{l,g} = c_{l,g,\mathrm{in}}\nabla A_{l,g}, \qquad \widetilde{\nabla}B_{l,g} = c_{l,g,\mathrm{out}}\nabla B_{l,g}. \tag{19}$$

Let $\widetilde{\nabla}M_{l,\mathrm{in}}$ be the row-block concatenation of $\{\widetilde{\nabla}A_{l,g}\}_{g=1}^{G}$, and let $\widetilde{\nabla}M_{l,\mathrm{out}}$ be the column-block concatenation of $\{\widetilde{\nabla}B_{l,g}\}_{g=1}^{G}$.

3.2.2 Role/Depth Matrix Step

The second control applies the matrix optimizer step to the scaled gradients. It assigns different depth trends to the input and output roles, runs AdamW on both matrices, and activates a transient Muon substep early in training. In the reported configuration, the AdamW multiplier depends on role and depth; the pressure and activity memories enter this control through the Muon attenuation factor. Let $\delta_l = (l-1)/(L-1)$ for $L > 1$ and $\delta_l = 1/2$ otherwise. The branch schedules are

$$\begin{aligned}
\varrho_{\mathrm{in}}(l) &= \mathrm{clip}(1 - 0.50(\delta_l - 0.5), 0.55, 1.40), \\
\varrho_{\mathrm{out}}(l) &= \mathrm{clip}(1 + 1.00(\delta_l - 0.5), 0.55, 1.40), \\
s_{l,\sigma}^{\mathrm{Adam}} &= \mathrm{clip}(3.0\max\{0.10, 1 + 1.20(\varrho_\sigma(l) - 1)\}, 0.40, 4.0), \\
\Psi_{l,t} &= \mathrm{clip}\left(G^{-1}\sum_{g=1}^{G}|\pi_{l,g}|, 0, 1.50\right), \\
\Omega_{l,t} &= \mathrm{ReLU}\left(\frac{G^{-1}\sum_{g=1}^{G}\alpha_{l,g} - 0.05}{0.45}\right), \\
\psi_{l,t} &= \mathrm{clip}(\exp[-0.30\Psi_{l,t} - 0.65\Omega_{l,t}], 0.10, 1.15), \\
\omega_t^{\mu} &= \mathrm{smoothstep}(0.02, 0.12, \xi_t)[1 - \mathrm{smoothstep}(0.20, 0.36, \xi_t)], \\
\mu_{l,\sigma,t} &= \mathrm{clip}(0.75\,\omega_t^{\mu}\varrho_\sigma(l)\psi_{l,t}, 0, 0.75).
\end{aligned} \tag{20}$$

Here $s_{l,\sigma}^{\mathrm{Adam}}$ is the role/depth AdamW multiplier. The aggregate pressure $\Psi_{l,t}$ and activity $\Omega_{l,t}$ form the Muon attenuation $\psi_{l,t}$, so the active Muon fraction $\mu_{l,\sigma,t}$ is largest during the early Muon window and shrinks when the detached summaries indicate large role imbalance or large rational-coefficient activity. For $M_{l,\mathrm{in}} = A_l$ and $M_{l,\mathrm{out}} = B_l$, let η_t be the base learning rate and let $s_\mu = 1.0$ be the evaluated Muon learning-rate scale. MatrixPolicy applies an AdamW matrix step followed by any active Muon substep. The gate $\mu_{l,\sigma,t}$ scales the two optimizer learning rates; AdamW and Muon

keep separate states and are applied sequentially:

$$\begin{aligned} \widetilde{M}_{l,\sigma}^{t+1} &= \mathrm{AdamW}_{\eta_t s_{l,\sigma}^{\mathrm{Adam}}(1-\mu_{l,\sigma,t})}(M_{l,\sigma}^{t}; \widetilde{\nabla} M_{l,\sigma}^{t}), \\ M_{l,\sigma}^{t+1} &= \mathrm{Muon}_{\eta_t s_\mu \mu_{l,\sigma,t}}(\widetilde{M}_{l,\sigma}^{t+1}; \widetilde{\nabla} M_{l,\sigma}^{t}), \end{aligned} \tag{21}$$

The AdamW step uses the training-protocol betas, weight decay, and learning-rate schedule in Appendix A.1. The Muon substep is the `torch.optim.Muon` update on two-dimensional RLB matrices with momentum 0.95, five Newton–Schulz steps, and match-RMS learning-rate adjustment (Liu et al., 2025); it becomes inactive when all $\mu_{l,\sigma,t}$ decay to zero.

3.2.3 Reciprocal Pair Rescaling

The third control chooses a bounded positive representative of the same RLB matrix pair after the matrix update. It is implemented by the surrounding transport wrapper and is evaluated every five optimizer steps after the AdamW/Muon branch; fixed-grid probe gains are refreshed every ten optimizer steps and reused between refreshes. It consumes the probe gains, the post-step log ratio $\gamma_{l,g}$, and the pressure/activity memories. The target combines the probe-gain balance coordinate derived in Appendix C.2 with a bounded pressure offset:

$$\begin{aligned} \bar{\pi}_{l,g} &= \mathrm{clip}(\pi_{l,g}, -1.25, 1.25), \\ \tau_{l,g} &= \frac{1}{2}\left(\log \hat{d}_{l,g}^{\mathrm{probe}} - \log \hat{o}_{l,g}^{\mathrm{probe}}\right) + 0.25\,\bar{\pi}_{l,g}, \\ \bar{\alpha}_{l,g} &= \mathrm{clip}(\alpha_{l,g}, -1.25, 1.25), \\ \chi_{l,g}^{\mathrm{rs}} &= \exp(0.10\,\bar{\alpha}_{l,g}), \qquad \text{(rescale-speed factor)} \\ s_{l,t} &= 0.50\ \mathrm{smoothstep}(0.03, 0.35, \xi_t)\,\mathrm{clip}(1 + 0.15(\delta_l - 0.5), 0.75, 1.25). \end{aligned} \tag{22}$$

The current log ratio $\gamma_{l,g}$ is measured after the matrix step. The clipped log step and reciprocal update are

$$\begin{aligned} \ell_{l,g,t} &= \mathrm{clip}\left(\frac{1}{2}[\tau_{l,g} - \gamma_{l,g}] s_{l,t} \chi_{l,g}^{\mathrm{rs}}, -0.030, 0.030\right), \\ A_{l,g} &\leftarrow e^{\ell_{l,g,t}} A_{l,g}, \qquad B_{l,g} \leftarrow e^{-\ell_{l,g,t}} B_{l,g}. \end{aligned} \tag{23}$$

The first term of $\tau_{l,g}$ equalizes derivative and response probe gains in the positive log coordinate, the pressure term shifts the target within a bounded range, and χ^{rs} changes the clipped rescale speed. Larger $\alpha_{l,g}$ accelerates this representative-balancing move, while the earlier gradient gate and Muon attenuation use the same activity contrast to reduce gradient-driven matrix motion. Because equation 22 uses floored RMS values, equation 23 is exact in the nonzero, unfloored idealization described by equation 9; near the RMS floor it is a clipped balance step. Optimizer-state buffers are scaled covariantly: first-moment-like entries for $A_{l,g}$ use $e^{\ell_{l,g,t}}$, first-moment-like entries for $B_{l,g}$ use $e^{-\ell_{l,g,t}}$, and square-average entries use the corresponding squared scales.

4 Experiments

4.1 Fixed-Scale Target-Arrival Study

The fixed-scale study uses the same 12-layer, width-768 causal Transformer on five corpora: DCLM/DataComp-LM (Li et al., 2024), FineWeb-Edu and FineWeb (Penedo et al., 2024), Dolma-sample (Soldaini et al., 2024), and C4 (Raffel et al., 2020). The 100M-token budget uses 3050 optimizer steps and the 300M-token budget uses 9150 optimizer steps, with three seeds per method/dataset pair. The main comparison keeps the architecture and data fixed while changing the activation and optimizer used in the nonlinear sublayer: MatrixPolicy is compared with SiLU controls and with RLB-only controls, where RLB-only uses the same RLB activation but no MatrixPolicy updates. Shared optimizer, evaluation-cadence, and throughput-measurement details are reported in Appendix A.1; broader optimizer comparisons beyond AdamW and Muon are summarized in Table 3.

Table 1: Target-arrival efficiency for the fixed 12-layer, width-768 Transformer. RLB denotes the same rational latent-basis activation without MatrixPolicy. At the hardest validation-loss target reached by every listed method in all three seeds, MatrixPolicy is always fastest; each token-saving column reports MatrixPolicy's savings relative to the method named below it.

			MatrixPolicy token savings (%)				Time to target (min)				
Budget	Dataset	Target loss	vs. SiLU AdamW	vs. SiLU Muon	vs. RLB AdamW	vs. RLB Muon	Matrix Policy	SiLU AdamW	SiLU Muon	RLB AdamW	RLB Muon
100M	DCLM	4.55	20.7	32.4	20.7	34.3	12.8	20.8	26.0	14.3	19.0
	FineWeb-Edu	4.40	19.5	29.5	20.2	29.5	12.9	20.3	24.4	14.2	18.0
	FineWeb	4.60	22.8	32.1	21.5	32.1	12.6	16.4	20.2	16.2	20.1
	Dolma	4.60	22.0	34.0	22.7	34.9	12.8	17.6	22.0	17.3	22.2
	C4	4.60	20.7	29.2	19.3	28.7	11.6	14.3	17.3	14.9	18.8
300M	DCLM	4.10	23.3	8.3	24.0	7.0	39.6	53.5	45.9	54.7	45.6
	FineWeb-Edu	3.85	22.8	6.7	22.6	4.8	40.0	59.9	47.9	52.3	50.0
	FineWeb	4.10	23.1	7.1	22.4	5.3	40.8	61.2	46.4	55.6	46.6
	Dolma	3.95	22.3	9.8	22.3	6.5	44.4	49.3	52.3	50.5	47.4
	C4	4.00	26.4	8.5	25.3	6.5	41.8	64.7	55.5	59.4	51.9

Table 1 is the primary empirical result. It asks: for the hardest common validation-loss target reached by every listed method in all three seeds, what fraction of target-arrival tokens does MatrixPolicy save, and how many measured target-arrival minutes does each method require? Across all five datasets under the 100M-token budget, MatrixPolicy saves 19.5–22.8% of the SiLU with AdamW target-arrival tokens and 29.2–34.0% of the SiLU with Muon target-arrival tokens. Under the 300M-token budget, after longer training compresses endpoint gaps, MatrixPolicy still saves 22.3–26.4% versus SiLU with AdamW and 6.7–9.8% versus SiLU with Muon, with lower target-arrival time in every listed comparison.

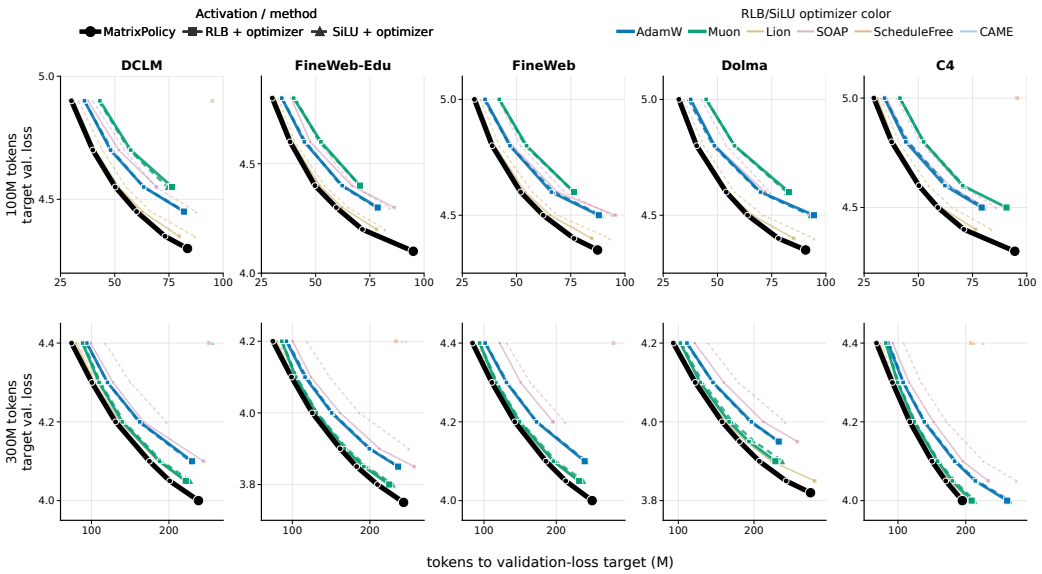

Figure 3: MatrixPolicy shifts target arrival left across the completed fixed-scale study. Rows show the 100M- and 300M-token budgets, and columns show datasets. Each point is the mean token count needed to reach one validation-loss target in all three seeds; connected points trace one method from easier to harder targets. Black circles are MatrixPolicy. For every optimizer color, solid squares are RLB with that optimizer and dashed triangles are SiLU with the same optimizer. Lower targets are harder, and the hard-target timing and token-saving percentages are quantified in Table 1.

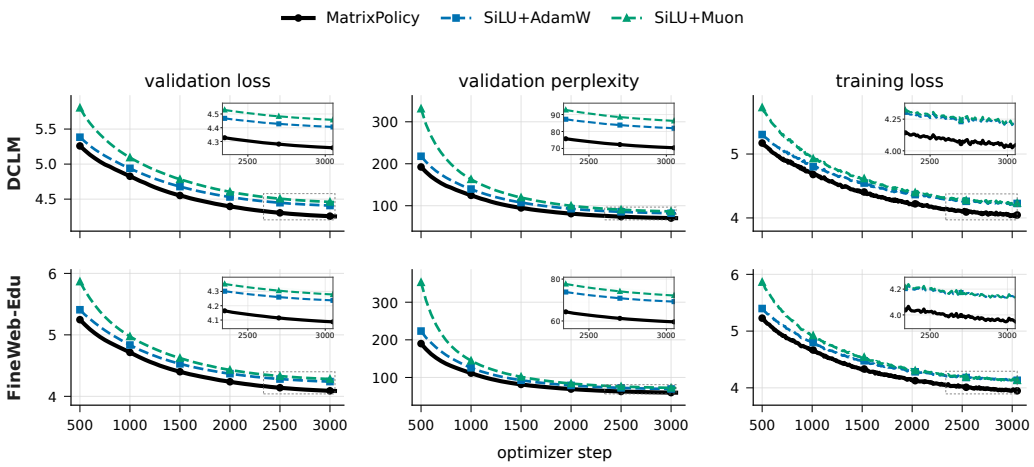

Figure 4: Representative 100M-token-budget training dynamics on DCLM and FineWeb-Edu. Rows show datasets and columns show validation loss, validation perplexity, and training loss. Lines are seed means over three runs and bands show one sample standard deviation. This figure uses the two standard SiLU baselines so that the early target-arrival effect in Table 1 remains visually legible; inset windows zoom the final-quarter saturation region.

The 100M-token-budget curves show why target arrival is the right main readout. MatrixPolicy separates early in validation loss and validation perplexity while keeping training loss competitive. A long fixed budget eventually compresses many activation/optimizer endpoints, especially under the 300M-token budget, so the paper emphasizes the number of tokens and minutes needed to reach the same useful validation-loss levels.

4.2 TODO: Larger-Scale Training Results

TODO: Insert the larger-scale training result here.

4.3 TODO: Component Ablations

TODO: Insert the component ablation study here.

5 Conclusion

RLB turns the nonlinear Transformer sublayer into an optimizer-visible interface by exposing grouped normalized coordinates, trainable groupwise rational responses, and an idealized positive matrix-pair relation used as a bounded balancing coordinate. MatrixPolicy uses that interface to update the RLB input and output matrices with role-aware statistics, while rational coefficients and non-RLB-matrix parameters use AdamW. The completed 100M-token and 300M-token runs show that MatrixPolicy reaches common validation losses with fewer tokens and lower measured target-arrival time than the listed SiLU and RLB controls. [todo: Complete this conclusion after the larger-scale and ablation results are added.]

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A Experiment Appendix

This appendix gives the fixed-scale protocol behind Section 4.1, the full training curves for the five corpora, and the broader optimizer endpoint comparison used to contextualize the main SiLU and RLB controls.

A.1 Fixed-Scale Target-Arrival Study

A.1.1 Model and Data Pipeline

The fixed-scale experiments use a custom causal decoder Transformer. The block uses a LLaMA-style pre-normalized layout with RMSNorm before attention and before the nonlinear sublayer, bias-free query/key/value and output attention projections, rotary position embeddings, causal scaled-dot-product attention, residual attention and nonlinear-sublayer updates, a final RMSNorm, and a tied token-embedding/output head (Zhang & Sennrich, 2019; Su et al., 2021; Touvron et al., 2023). The fixed model has 12 layers, width 768, 12 heads of dimension 64, and context length 256.

The SiLU controls use a SwiGLU-like gated nonlinear sublayer with bias-free matrices $W_g, W_v \in \mathbb{R}^{768\times 2048}$ and $W_o \in \mathbb{R}^{2048\times 768}$, computing $(\mathrm{SiLU}(xW_g) \odot xW_v)W_o$. MatrixPolicy and RLB-only configurations use a matched single-branch matrix pair. An input matrix maps width 768 to 3072 latent coordinates, the coordinates are partitioned into groups and transformed by the real-pole-free P5/Q4 rational response, and an output matrix maps the 3072 coordinates back to width

768. With the default group size 256, this gives $G = 12$ groups of width $m = 256$; the RMS floor is $\epsilon_{\text{rms}} = 10^{-6}$. This is the RLB configuration used by the main MatrixPolicy and RLB-control results.

Tokenization uses the GPT-2 tokenizer through the Hugging Face tokenizer interface, without adding tokenizer-specific special tokens; an EOS or pad token is appended between documents. Training samples random contiguous blocks of length 257 and forms next-token prediction pairs of length 256. The DCLM, FineWeb-Edu, FineWeb, Dolma-sample, and C4 runs use the text field specified in the manifests, with validation slices drawn from held-out token offsets of the same corpus stream.

A.1.2 Training Protocol

All runs use batch size 16, gradient accumulation 2, and therefore 32768 global tokens per optimizer step. The 100M-token budget consists of 3050 optimizer steps; the 300M-token budget consists of 9150 optimizer steps. Each method/dataset pair is run with seeds 1337, 2027, and 3407. Validation is performed every 50 optimizer steps, so token-to-target arrivals are quantized in increments of 1.64M tokens.

The MatrixPolicy rows in the main table use the RLB activation, whose active nonlinearity is the groupwise P5/Q4 response in equations 3–5. The evaluated configuration uses AdamW as the backbone optimizer and disables the separate rational-coefficient optimizer. It enables MatrixPolicy group redistribution with gain/pressure/activity strengths $0.20/0.10/0.20$, schedule window 0.02–0.30, and group multiplier clip $[0.75, 1.35]$. The RLB matrix pair then uses the role/depth-scaled AdamW branch and the early Muon substep from Section 3.2. The bounded RLB pair-rescale wrapper runs every five optimizer steps with covariant optimizer-state scaling. Rational coefficients, attention weights, embeddings, normalization parameters, and ordinary Transformer weights use AdamW in this configuration.

Target-arrival minutes in Table 1 are computed seed-wise from the first validation event that reaches the shared target and the same run's measured seconds per optimizer step. The per-step time is the training-script wall time recorded in the run summary divided by completed optimizer steps for that run; target arrivals are therefore measured with the same evaluation cadence and timing source across methods.

A.1.3 Learning-Rate and Weight-Decay Sensitivity

The main comparison fixes one optimizer hyperparameter setting per method. To test whether the target-arrival pattern survives nearby choices, we run a paired 100M-token grid on DCLM, FineWeb-Edu, and C4 with learning rates $\{10^{-4}, 2 \cdot 10^{-4}, 3 \cdot 10^{-4}, 5 \cdot 10^{-4}\}$ and weight decays $\{0, 0.05, 0.10, 0.20\}$. Table 2 reports the same target-arrival quantities as Table 1. All-grid rows use the hardest 0.05-spaced target reached by every listed method in all 16 cells; main-target rows reuse the harder targets from Table 1 where all listed methods reach them. MatrixPolicy saves tokens and lowers mean target-arrival minutes in both readouts, and it has lower final validation loss in all 48 matched endpoint comparisons.

Table 2: Learning-rate and weight-decay sensitivity in the 100M-token 12-layer setting. Values are matched-cell means using the same target-arrival computation as Table 1.

				MatrixPolicy token savings (%)		Time to target (min)		
Dataset	Target rule	Target loss	Cells	vs. SiLU AdamW	vs. SiLU Muon	Matrix Policy	SiLU AdamW	SiLU Muon
DCLM	All-grid	5.25	16	17.5	44.0	5.0	5.3	9.9
	Main target	4.55	8	13.6	19.7	10.4	10.5	12.6
FineWeb-Edu	All-grid	5.20	16	18.1	38.1	5.1	5.5	9.3
	Main target	4.40	8	14.4	19.1	10.2	10.4	12.2
C4	All-grid	5.40	16	17.3	40.0	4.8	5.1	8.6
	Main target	4.60	8	14.8	19.2	9.9	10.1	11.9

A.1.4 SUPPORTING FIXED-SCALE CURVES

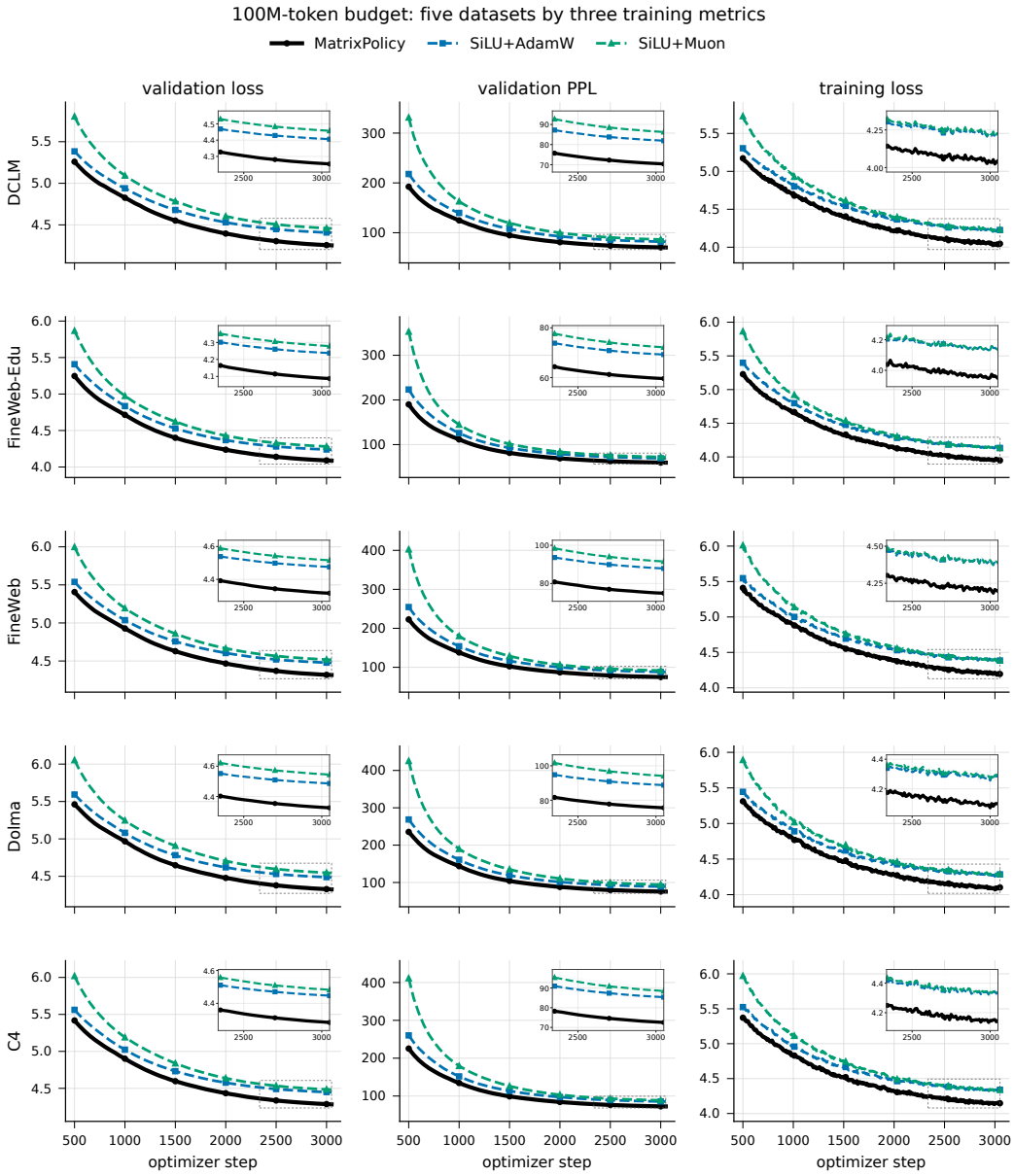

Figure 5: 100M-token-budget dynamics across all five datasets. Rows are datasets and columns are validation loss, validation perplexity, and training loss. Lines are seed means over three runs and bands show one sample standard deviation. Insets zoom the final-quarter saturation region in each panel.

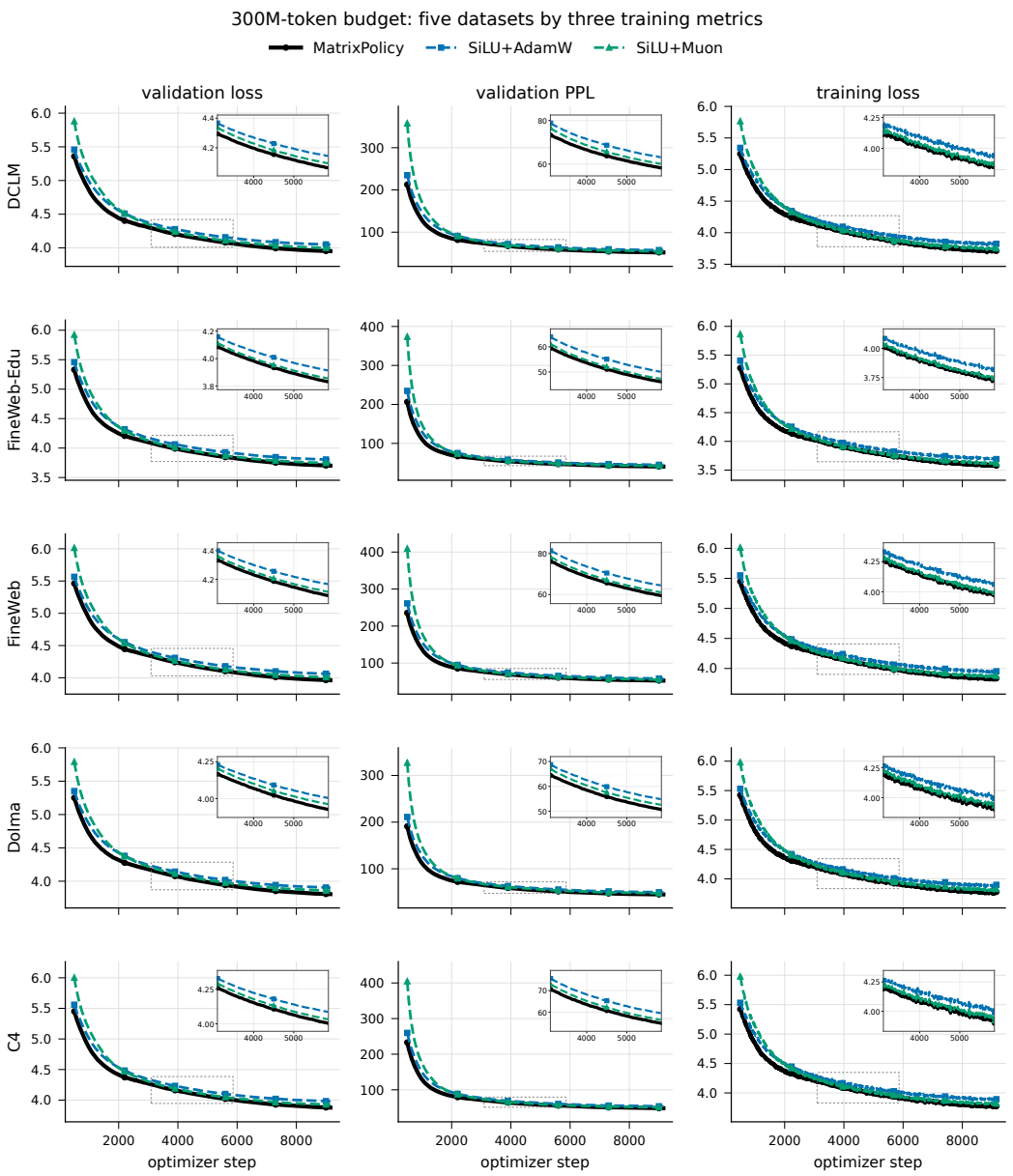

Figure 6: 300M-token-budget dynamics across all five datasets. Rows are datasets and columns are validation loss, validation perplexity, and training loss. Lines are seed means over three runs and bands show one sample standard deviation. Insets zoom the middle training window where early saturation differences are most visible.

A.1.5 Broad Optimizer Endpoint Table

The fixed-scale package also includes the broader optimizer endpoints referenced from Section 4.1. These rows use the same model, datasets, seeds, token budgets, and validation schedule as the main comparison, while replacing the nonlinear-sublayer optimizer according to the table entry.

Table 3: Final validation loss for the broader optimizer sweep on the fixed 12-layer, width-768 language model. Values are mean ± sample standard deviation over three seeds; lower is better. RLB rows use the same rational latent-basis activation without MatrixPolicy.

Budget	Method	DCLM	FineWeb-Edu	FineWeb	Dolma	C4
100M tokens	MatrixPolicy	4.254±0.006	4.087±0.010	4.316±0.013	4.325±0.005	4.284±0.020
	SiLU + AdamW	4.406±0.010	4.237±0.009	4.476±0.010	4.486±0.001	4.447±0.016
	SiLU + Lion	4.318±0.007	4.149±0.009	4.383±0.008	4.388±0.005	4.354±0.016
	SiLU + Muon	4.457±0.013	4.279±0.024	4.516±0.026	4.544±0.011	4.482±0.023
	SiLU + SOAP	4.416±0.004	4.263±0.022	4.484±0.011	4.499±0.004	4.458±0.015
	SiLU + ScheduleFree	4.902±0.011	4.826±0.007	5.014±0.019	5.049±0.015	5.006±0.016
	SiLU + CAME	5.011±0.014	4.921±0.012	5.133±0.013	5.165±0.019	5.123±0.018
	RLB + AdamW	4.405±0.008	4.239±0.008	4.472±0.014	4.488±0.003	4.441±0.016
	RLB + Lion	4.295±0.008	4.136±0.008	4.363±0.011	4.362±0.007	4.327±0.016
	RLB + Muon	4.467±0.012	4.283±0.028	4.516±0.006	4.548±0.015	4.476±0.010
	RLB + SOAP	4.436±0.044	4.275±0.024	4.689±0.336	4.518±0.036	4.446±0.019
	RLB + ScheduleFree	4.887±0.005	4.796±0.013	4.999±0.019	5.036±0.013	4.984±0.016
	RLB + CAME	5.014±0.009	4.915±0.006	5.134±0.016	5.151±0.017	5.111±0.016
300M tokens	MatrixPolicy	3.952±0.028	3.702±0.021	3.962±0.008	3.806±0.007	3.878±0.014
	SiLU + AdamW	4.049±0.027	3.803±0.018	4.061±0.010	3.904±0.009	3.981±0.013
	SiLU + Lion	3.993±0.023	3.744±0.021	4.001±0.008	3.848±0.009	3.921±0.011
	SiLU + Muon	3.997±0.030	3.745±0.017	4.007±0.013	3.858±0.010	3.925±0.013
	SiLU + SOAP	4.096±0.030	3.863±0.020	4.114±0.010	3.957±0.009	4.035±0.011
	SiLU + ScheduleFree	4.366±0.030	4.156±0.024	4.398±0.011	4.215±0.005	4.316±0.011
	SiLU + CAME	4.368±0.023	4.150±0.021	4.406±0.019	4.249±0.027	4.330±0.015
	RLB + AdamW	4.050±0.030	3.802±0.021	4.060±0.011	3.905±0.008	3.979±0.010
	RLB + Lion	3.989±0.029	3.742±0.021	3.996±0.011	3.841±0.008	3.913±0.014
	RLB + Muon	3.991±0.026	3.737±0.019	3.999±0.011	3.848±0.005	3.919±0.015
	RLB + SOAP	4.061±0.033	3.824±0.013	4.108±0.032	3.927±0.013	4.002±0.019
	RLB + ScheduleFree	4.361±0.034	4.142±0.022	4.386±0.008	4.212±0.011	4.308±0.007
	RLB + CAME	4.450±0.043	4.225±0.036	4.480±0.006	4.267±0.041	4.365±0.058

A.2 TODO: Larger-Scale Training Results

TODO: Insert larger-scale training details here.

A.3 TODO: Component Ablations

TODO: Insert component ablation details here.

B Rational Latent Basis Calculus

All calculations in this section are local to one layer and one group unless indices are shown explicitly.

B.1 Scalar Response Derivatives

The evaluated response is the groupwise P5/Q4 function defined in the method. Writing

$$\phi_1(u) = |u|, \quad \phi_2(u) = u^2, \quad \phi_3(u) = |u|^3, \quad \phi_4(u) = u^4, \qquad Q(u) = 1 + \sum_{j=1}^{4} |b_j| \phi_j(u), \tag{24}$$

gives $Q(u) \geq 1$ on the real line. Thus the scalar response has no real pole, while its numerator and denominator coefficients remain explicit trainable coordinates. Away from $u = 0$,

$$R'(u) = \frac{P'(u)Q(u) - P(u)Q'(u)}{Q(u)^2}, \tag{25}$$

$$P'(u) = \sum_{i=1}^{5} i a_i u^{i-1}, \qquad Q'(u) = |b_1| \operatorname{sign}(u) + 2|b_2| u + 3|b_3| u|u| + 4|b_4| u^3. \tag{26}$$

The derivative-gain summaries use the same autodiff convention as the code, $\operatorname{sign}(0) = 0$ at absolute-value kinks. The coefficient features are

$$\frac{\partial R(u)}{\partial a_i} = \frac{u^i}{Q(u)}, \qquad \frac{\partial R(u)}{\partial b_j} = -\frac{P(u)}{Q(u)^2} \operatorname{sign}(b_j) \phi_j(u), \tag{27}$$

with $\operatorname{sign}(0) = 0$ at $b_j = 0$. Consequently the three summary families in the method have direct scalar origins: $R(u)$ for response gain, $R'(u)$ for derivative gain, and coefficient gradients through the features in equation 27. For the evaluated P5/Q4 response, $\Theta^R_{l,g} = \{a_{l,g,\cdot}, b_{l,g,\cdot}\}$ in equation 12; if another implementation adds further trainable response-coordinate blocks, the same block-average definition extends to those blocks.

B.2 RMS-Normalized Group Map and Positive Pair Coordinate

For a group vector $z \in \mathbb{R}^m$, define

$$r(z) = \sqrt{m^{-1}\|z\|_2^2 + \epsilon_{\mathrm{rms}}}, \qquad u(z) = z/r(z), \qquad h(z) = r(z)R(u(z)), \tag{28}$$

where R acts coordinatewise. On the domain $r(z) > 0$, the radius and normalized-coordinate differentials are

$$\mathrm{d}r = \frac{z^\top \mathrm{d}z}{mr} = \frac{u^\top \mathrm{d}z}{m}, \qquad \mathrm{d}u = \frac{1}{r}\left(I - \frac{uu^\top}{m}\right)\mathrm{d}z. \tag{29}$$

The factor $I - uu^\top/m$ is the exact differential factor of the implemented normalized coordinate; it becomes the tangent projector to the RMS sphere in the unfloored case. Away from scalar-response kinks, the group Jacobian is

$$J_h(z) := \frac{\partial h}{\partial z} = R(u)\frac{u^\top}{m} + \operatorname{diag}(R'(u))\left(I - \frac{uu^\top}{m}\right). \tag{30}$$

The Jacobian in equation 30 separates the radial response term from the normalized-coordinate derivative term. Response RMS therefore serves as an output-role signal, and derivative RMS serves as an input-role signal. Both remain proxies: response RMS omits the radius, while derivative RMS omits the radial Jacobian term, the output matrix, and the upstream gradient.

The same normalization gives the positive matrix-pair coordinate used by the wrapper. If $\epsilon_{\mathrm{rms}} = 0$, $z \neq 0$, and $\lambda > 0$, then

$$r(\lambda z) = \lambda r(z), \qquad u(\lambda z) = u(z), \qquad h(\lambda z) = \lambda h(z). \tag{31}$$

Scaling $A_{l,g}$ by λ and the matching block $B_{l,g}$ by λ^{-1} leaves that group contribution unchanged, giving equation 9 groupwise. With the implemented floor, write

$$r_\epsilon(z) = \sqrt{m^{-1}\|z\|_2^2 + \epsilon_{\mathrm{rms}}}, \qquad u_\epsilon(z) = z/r_\epsilon(z), \qquad h_\epsilon(z) = r_\epsilon(z)R(u_\epsilon(z)). \tag{32}$$

For $\rho^2 = m^{-1}\|z\|_2^2$,

$$u_\epsilon(\lambda z) = \kappa(\lambda, z)u_\epsilon(z), \qquad \kappa(\lambda, z) = \frac{\lambda\sqrt{\rho^2 + \epsilon_{\mathrm{rms}}}}{\sqrt{\lambda^2\rho^2 + \epsilon_{\mathrm{rms}}}}. \tag{33}$$

If L_R is a Lipschitz constant of the scalar response on an interval containing the coordinates of $u_\epsilon(z)$ and $\kappa(\lambda, z)u_\epsilon(z)$, then after the reciprocal output compensation,

$$\begin{aligned}\left\|\lambda^{-1}h_\epsilon(\lambda z) - h_\epsilon(z)\right\|_2 \le{}& \left|\frac{r_\epsilon(\lambda z)}{\lambda} - r_\epsilon(z)\right| \left\|R(u_\epsilon(z))\right\|_2 \\ &+ \frac{r_\epsilon(\lambda z)}{\lambda} L_R |\kappa(\lambda, z) - 1| \left\|u_\epsilon(z)\right\|_2 .\end{aligned} \tag{34}$$

Both terms vanish when $\epsilon_{\mathrm{rms}} = 0$. Under bounded rescaling $|\log\lambda| \le c$, the deviation is controlled when $e^{2c}\epsilon_{\mathrm{rms}}/\rho^2$ is small and the response is bounded with a bounded derivative on the interval above. Near the RMS floor, the coordinate is therefore a clipped representative direction, with exact balance symmetry recovered in the unfloored block.

C MatrixPolicy Update Geometry

C.1 Detached Summaries as Matrix-Role Proxies

For one training example, let $\bar{y}_l = \nabla_{y_l}\mathcal{L}$ and $J_{l,g} = \partial h_{l,g}/\partial z_{l,g}$ from equation 30. The adjacent matrix gradients factor as

$$\nabla_{B_{l,g}}\mathcal{L} = \bar{y}_l h_{l,g}^\top, \qquad \nabla_{A_{l,g}}\mathcal{L} = \left(J_{l,g}^\top B_{l,g}^\top \bar{y}_l\right) x_l^\top. \tag{35}$$

The factorization in equation 35 separates the two local role signals. The output-role gradient depends on the realized feature $h_{l,g}$, whereas the input-role gradient depends on the local RLB Jacobian before propagation to x_l. The response RMS and derivative RMS in equation 15 therefore serve as local output- and input-role gain proxies, with data distribution, radius factors, matrix orientation, and upstream gradients supplied by the full training gradient.

The pressure memories in equation 13 summarize post-global-clipping, pre-group-policy gradients as update-magnitude proxies. Their signed contrast is $\pi_{l,g} = \log q_{l,g}^{\text{in}} - \log q_{l,g}^{\text{out}}$. Along the exact positive pair direction of the unfloored fixed-curve map, reciprocal infinitesimal rescaling has zero data-loss directional derivative. For nonzero unfloored W and a small update $W^+ = W - \eta\mathcal{G}$ with $\eta\|\mathcal{G}\|_F < c\|W\|_F$ for fixed $c < 1$,

$$\log \operatorname{rms}(W^+) - \log \operatorname{rms}(W) = -\eta\frac{\langle \mathcal{G}, W\rangle_F}{\|W\|_F^2} + O\left(\eta^2 \frac{\|\mathcal{G}\|_F^2}{\|W\|_F^2}\right), \tag{36}$$

and by Cauchy–Schwarz,

$$\left|\frac{\langle \mathcal{G}, W\rangle_F}{\|W\|_F^2}\right| \le \frac{\|\mathcal{G}\|_F}{\|W\|_F} = \frac{\operatorname{rms}(\mathcal{G})}{\operatorname{rms}(W)}. \tag{37}$$

The implemented p^{in} and p^{out} replace the last ratio by floored RMS proxies. The floor regularizes the ratio near zero, while the clips in the method turn their log contrasts into bounded controls. Thus $\pi_{l,g}$ identifies which role currently has the larger EMA of the relative update proxy, and $\alpha_{l,g}$ identifies when raw rational-coefficient gradient energy is large relative to adjacent matrix-pressure memories.

C.2 Geometry of the Three MatrixPolicy Controls

The group-gradient policy in equation 18 acts in log space. Let

$$\bar{d}_l = \frac{1}{G}\sum_{j=1}^{G} \log \hat{d}_{l,j}^{\text{live}}, \qquad \bar{o}_l = \frac{1}{G}\sum_{j=1}^{G} \log \hat{o}_{l,j}^{\text{live}}. \tag{38}$$

Before clipping and activity damping, the gain component satisfies

$$\log c_{l,g,\text{in}}^{\text{gain}} = -0.20\phi_t(\log \hat{d}_{l,g}^{\text{live}} - \bar{d}_l), \qquad \log c_{l,g,\text{out}}^{\text{gain}} = -0.20\phi_t(\log \hat{o}_{l,g}^{\text{live}} - \bar{o}_l). \tag{39}$$

Thus groups with above-geometric-mean sampled derivative or response gain receive smaller gradient multipliers in the corresponding role, while below-mean groups receive larger multipliers. The pressure factor has the role signs

$$\log c_{l,g,\text{in}}^{\text{pressure}} = -0.10\phi_t\,\tilde{\pi}_{l,g}, \qquad \log c_{l,g,\text{out}}^{\text{pressure}} = +0.10\phi_t\,\tilde{\pi}_{l,g}, \tag{40}$$

before geometric recentering. Hence larger input-role pressure decreases the input-side log scale and increases the output-side log scale relative to groups with smaller pressure contrast. Recentring by the group geometric mean gives

$$\widehat{c}_{l,g,\sigma} = \frac{c_{l,g,\sigma}}{\operatorname{geomean}_{j=1}^{G} c_{l,j,\sigma}}, \qquad \frac{1}{G}\sum_{g=1}^{G} \log \widehat{c}_{l,g,\sigma} = 0. \tag{41}$$

Together, the gain, pressure, and recentering equations make the first control a bounded redistribution of role-specific update scale. The role-wise mean log multiplier is zero before clipping, and all nonzero directional information comes from the current gradient.

The second control allocates the matrix learning-rate budget between a coordinate-adaptive AdamW direction and a transient matrix-normalized Muon direction. Let $D^{\mathrm{Adam}}_{l,\sigma,t}$ be the AdamW matrix direction and let $D^{\mu}_{l,\sigma,t}$ be the Muon direction. To first order in η_t, suppressing weight decay and the $O(\eta_t^2)$ term from sequential composition, the branch in equation 21 has the allocation

$$\Delta M_{l,\sigma} = -\eta_t s^{\mathrm{Adam}}_{l,\sigma}(1-\mu_{l,\sigma,t})D^{\mathrm{Adam}}_{l,\sigma,t} - \eta_t s_\mu \mu_{l,\sigma,t} D^{\mu}_{l,\sigma,t} + O(\eta_t^2). \tag{42}$$

For the compact singular value decomposition of the Muon momentum matrix $H_{l,\sigma,t} = U\Sigma V^\top$, define the polar direction

$$\mathcal{P}(H_{l,\sigma,t}) = UV^\top. \tag{43}$$

This polar direction is the solution of the spectral-norm constrained linearized descent problem,

$$\mathcal{P}(H) \in \operatorname*{arg\,max}_{\|D\|_2 \le 1} \langle D, H\rangle_F, \qquad \langle \mathcal{P}(H), H\rangle_F = \|H\|_*, \qquad \mathcal{P}(aH) = \mathcal{P}(H) \quad (a>0). \tag{44}$$

Muon applies the finite Newton–Schulz approximation supplied by `torch.optim.Muon`, with momentum and the match-RMS learning-rate adjustment used in the implementation. The branch supplies an early matrix-normalized component whose singular-direction scale is equalized before the scalar learning-rate gate is applied. The gate in equation 20 is monotone in the two layer summaries. Inside the clipping interval,

$$\frac{\partial \log \psi_{l,t}}{\partial \Psi_{l,t}} = -0.30, \qquad \frac{\partial \log \psi_{l,t}}{\partial \Omega_{l,t}} = -0.65. \tag{45}$$

Thus the matrix-polar direction is strongest when role pressure and rational-coefficient activity are both small. The scheduled window ω^{μ}_t makes this component transient, while AdamW remains the persistent matrix update in the second control.

In the unfloored, unclipped pair-rescale model, define

$$\begin{aligned} \gamma_{l,g} &= \log \mathrm{rms}(A_{l,g}) - \log \mathrm{rms}(B_{l,g}), \\ \lambda_{l,g,t} &= s_{l,t}\chi^{\mathrm{rs}}_{l,g}, \\ \gamma^{+}_{l,g} = \gamma_{l,g} + 2\ell_{l,g,t} &= (1-\lambda_{l,g,t})\gamma_{l,g} + \lambda_{l,g,t}\tau_{l,g}. \end{aligned} \tag{46}$$

Because $0 \le s_{l,t} \le 0.625$ and $\chi^{\mathrm{rs}}_{l,g} \le \exp(0.125)$ under the method clips, $0 \le \lambda_{l,g,t} < 1$ before log-step clipping, so the representative coordinate contracts toward $\tau_{l,g}$. This model assumes nonzero matrices, unfloored RMS values, and an unclipped update; the implementation uses floored $\gamma_{l,g}$ and clipped $\ell_{l,g,t}$.

The target in equation 22 equalizes a local representative model that suppresses orientation, radius variation, upstream gradients, and floors. In that coordinate, input-role scale varies as $\hat{d}^{\mathrm{probe}}e^{-\gamma}$ and output-role scale varies as $\hat{o}^{\mathrm{probe}}e^{\gamma}$. Equalizing the two log scales gives

$$\gamma^{\star}_{l,g} = \frac{1}{2}\left(\log \hat{d}^{\mathrm{probe}}_{l,g} - \log \hat{o}^{\mathrm{probe}}_{l,g}\right). \tag{47}$$

The implemented target adds the clipped RMS-pressure bias $0.25\bar{\pi}_{l,g} \in [-0.3125, 0.3125]$, while $\chi^{\mathrm{rs}}_{l,g}$ changes only the representative-balancing speed. Together with Appendix B.2, these equations specify the approximation boundary of the pair-rescale move used in the method.